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HMT-KD: Lightweight deep learning model for ECG analysis using hierarchical multi-teacher knowledge distillation framework

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HIGHLIGHTS

- Proposing a novel hierarchical multi-teacher knowledge distillation framework.
- Enhanced Teacher Assistant training.
- Designing a compact and efficient student model.
- Employing a dual-path knowledge distillation strategy.
- Comprehensive evaluation on benchmark datasets.

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ABSTRACT

With the rapid expansion of intelligent healthcare systems, the demand for accurate yet lightweight deep learning models capable of real-time electrocardiogram (ECG) analysis has grown substantially. While traditional deep models achieved high diagnostic accuracy, their high computational resource requirement hinder integration into clinical decision support and wearable health-monitoring systems. To overcome this challenge, this study introduces a Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework, where three deep neural networks with different structure and model size collaboratively try to enhance and train a lightweight student model. At first, transfer learning technique along with fine tuning is employed for two high-capacity teacher networks, ResNet-18 with SENet and MobileNetV2, to act as advanced feature extractors. Then, a Teacher Assistant (TA) network, as an effective intermediary model, is trained using the Knowledge Distillation (KD) method based on the responses by both teachers. Finaly the lightweight student, which has similar architecture to the TA is trained under the joint supervision of the two teachers and the TA through two parallel learning pathways: 1) response-based KD from both primary teachers and the TA, and 2) feature-based KD from the intermediate layers of the TA. Experimental results on PTB-XL and Chapman datasets show that the student model reduces parameter count by over 100 times compared to the ResNet-18 +SENet teacher while maintaining high diagnostic performance 85.45% accuracy and 96.53% AUROC on PTB-XL and 94.42% accuracy and 98.86% AUROC on Chapman. These findings demonstrate the proposed framework's effectiveness in creating lightweight, accurate, and real-time ECG-based diagnostic models.

1. Introduction

Cardiovascular diseases claim approximately 17.9 million lives annually, positioning them among the foremost global health challenges [1]. Early detection and continuous monitoring of cardiac abnormalities are therefore crucial for improving patient outcomes and reducing healthcare burdens. Electrocardiography (ECG), as a non-invasive and clinically informative diagnostic tool, plays a vital role in detecting cardiac dysfunction. However, manual ECG interpretation requires expert knowledge and is subject to inter-observer variability, limited scalability, and reduced efficiency in high-throughput clinical environments. This challenge has catalyzed the pursuit of automated solutions capable of delivering rapid and consistent assessments.

Deep learning methods have recently garnered significant attention for the analysis of electrocardiogram (ECG) signals, providing novel avenues for uncovering clinically meaningful patterns [2]. ECG signals

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are challenging due to their complex temporal and spatial structures. Recurrent networks like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) address temporal dependencies [3,4], while One-Dimensional Convolutional Neural Networks (1D CNNs) effectively capture localized features, improving diagnostic accuracy [5–7]. More recently, Shi et al. [8] proposed a framework combining self-supervised generative and contrastive learning with 1D-VGGNet and 1D-ResNet to enhance ECG reconstruction and classification. Additionally, Strodthoff et al. [9], presented a benchmark on the PTB-XL dataset using advanced models like xResNet1D and Inception1D for standardized evaluation of ECG tasks. While 1D CNNs and recurrent networks can extract temporal features from ECG signals, they are less effective at capturing complex spatial and morphological patterns. Transforming ECG signals into two-dimensional representations allows the use of pre-trained 2D CNNs. This approach is particularly valuable in healthcare, where leveraging pre-trained models on large datasets such as ImageNet can mitigate the challenges of limited labeled data and improve generalizability.

Several studies [10–13] have shown that converting 1D ECG signals into 2D image-like forms and employing transfer learning with pre-trained convolutional networks significantly improved diagnostic accuracy and robustness. For example, continuous wavelet transform (CWT) with an optimized AlexNet has achieved high-accuracy cardiac signal classification [10]. Recurrence Plot (RP) visualization has enabled successful use of models like VGG-16, VGG-19, and AlexNet for detecting cardiac issues [11]. Additionally, combining Stockwell Transform (ST) and CWT with architectures such as Xception, VGG-19, DenseNet-121, and ResNetV2 has further enhanced classification performance [12].

Despite advances, many deep learning models remain complex and resource-heavy, limiting their use in real-time, portable, and wearable devices. To address this, knowledge transfer from twelve-lead ECG models to lightweight single-lead models has shown improved accuracy suitable for resource-constrained settings [14].Additionally, hybrid frameworks combining multi-task learning, multi-view learning, and knowledge distillation (KD) enhanced myocardial infarction detection accuracy while reducing complexity and improving generalizability [15]. In multi-label classification, multi-view and multi-scale networks with KD achieved faster processing and fewer parameters [16]. These advances suggest that knowledge distillation can enable accurate, efficient, and low-power medical systems like wearable ECG devices.

Despite these successes, most deep models remain computationally intensive and memory-demanding, limiting their real-time deployment in wearable or resource-constrained clinical devices. As healthcare systems move toward edge intelligence and ubiquitous monitoring, there is a critical need for lightweight deep learning models that maintain diagnostic accuracy while operating efficiently under limited computational budgets. In this article, an innovative Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework is proposed to efficiently train a lightweight student network using ECG signal images, targeting deployment on resource-constrained devices. These ECG signal images offer an intuitive visualization of waveform morphology, amplitude variations, and spatial patterns, facilitating more efficient feature learning. Furthermore, they are readily obtainable in real-world clinical environments, commonly provided as printed or digital ECG reports [17]. Unlike conventional KD methods that rely on a single teacher, which are often limited in capturing diverse feature representations and therefore may restrict the student model's diagnostic capability, the proposed framework employs two pre-trained high-capacity primary teacher networks ResNet-18 integrated with a SENet module and MobileNetV2 both capable of extracting rich and effective features to enhance the student's learning.

To further enhance the knowledge transfer process, a Teacher Assistant (TA) network is introduced, architecturally inspired by both primary teachers to better capture and comprehend the rich feature representations they have learned, thereby providing more effective guidance to the student model. The TA first learns as a student through response-based KD from the two teachers. This stage enriches its representational capacity, enabling it to serve as an effective bridge between the high-capacity teachers and the lightweight student. The student model has a structure similar to the TA network, and this design enables the effective identification and utilization of the rich intermediate feature layers learned by the TA. It integrates Depthwise Separable Convolutions (DSCs) to reduce computational cost while preserving essential signal characteristics, Residual Blocks (RBs) to facilitate the extraction of deep, high-level features and to mitigate the vanishing gradient problem during training, and a simplified Convolutional Block Attention Module (CBAM) to emphasize the most informative feature channels particularly those corresponding to key ECG components such as the P wave, QRS complex, and T wave. This architecture enables the student model to efficiently interpret meaningful intermediate features from the TA, thereby enhancing its diagnostic accuracy and overall performance.

It learns through two parallel learning paths: (1) response-based KD from the teachers and TA, and (2) feature-based KD from intermediate layers of the TA. These two paths help the lightweight student model benefit not only from the valuable knowledge in the teachers' outputs, but also from the rich intermediate features of the Teacher Assistant, which has a similar architecture to the student yet possesses much more informative representations. The proposed multi-stage, multi-path strategy enables the student network to preserve a highly compact architecture while achieving remarkable efficiency. Despite containing more than 100 times fewer parameters than the ResNet18 +SENet teacher model, the student consistently outperforms it across all evaluation metrics. This superior performance underscores the robustness, reliability, and practical relevance of the framework. Beyond computational efficiency, the lightweight architecture is carefully designed for deployment on low-power and portable devices, making it particularly suitable for real-world medical applications. By uniting precision, efficiency, and deployability, this approach highlights a transformative potential for clinical monitoring and portable healthcare systems.

The main contributions of this research are briefly summarized as follows:

- **Proposing a novel Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework:** Introducing two high-capacity teacher models, ResNet-18 enhanced with SENet and MobileNetV2, which are capable of extracting high-level features, along with an intermediate TA network.
- **Enhanced Teacher Assistant training:** The TA network is first trained through response-based KD from the primary teacher models, improving its representation capacity and preparing it to better guide the student model. Its architecture is designed based on the primary teacher networks to leverage their efficient modules and high-capacity feature extraction capabilities.
- **Designing a compact and efficient student model:** The student model, inspired by the architectures of the primary teachers and TA, employs DSC, RB, and a streamlined CBAM attention module (channel attention only), achieving high diagnostic accuracy with reduced computational cost and designed to run efficiently on resource-constrained devices.
- **Employing a dual-path knowledge distillation strategy:** The student learns via response-based distillation from both the primary teacher models and TA, as well as feature-based distillation from the intermediate layers of the TA, enhancing robustness and overall performance.
- **Comprehensive evaluation on benchmark datasets:** Extensive experiments on PTB-XL and Chapman demonstrate that the student model delivers competitive or superior diagnostic accuracy compared to the ResNet18 +SENet teacher model, while achieving over 100 times reduction in parameters, highlighting its suitability for real-time and resource-constrained environments.

The paper is structured as follows: Section 2 provides a review of related work and prior research on ECG analysis and knowledge distillation. Section 3 details the proposed HMT-KD framework. Section 4 presents the experimental setup, results, and analysis on benchmark ECG datasets. Finally, Section 5 concludes the study and outlines potential directions for future research.

2. Related work

ECG signal analysis has been a core area of interest in biomedical signal processing due to its diagnostic significance. However, the inherent structural complexity of ECG signals, their vulnerability to various noise sources, and the severe class imbalance in clinical datasets pose substantial challenges for conventional approaches. To address these limitations, deep learning-based methods particularly Recurrent Neural Networks (RNNs) and 1D-CNNs have gained considerable attention in recent years. These models have shown promising capabilities in automatic feature extraction and accurate classification of abnormal cardiac patterns [18–20].

RNNs are particularly effective in modeling the sequential nature of ECG signals, as they can capture long-range temporal dependencies [21]. For example, Gao et al. [3] applied an LSTM model to detect arrhythmias in highly imbalanced ECG datasets, while Lynn et al. [4] enhanced classification performance using a Bidirectional GRU architecture across various datasets. While models like RNNs are effective at capturing temporal dependencies, they often struggle to extract fine-grained local features. In contrast, 1D-CNNs have proven particularly effective for identifying cardiac abnormalities by capturing local signal patterns [22]. For example, Śmigiel et al. [5] and Lassoued and Ketata [23] successfully combined 1D-CNNs with R-peak detection algorithms such as Pan-Tompkins and Christov to improve ECG classification. Similarly, Jahmunah et al. [24] enhanced the interpretability of deep models by incorporating Gradient-weighted Class Activation Mapping (Grad-CAM). Recent works have also explored the integration of attention modules [6] and advanced architectures such as Siamese networks [25] aiming to improve feature extraction, reduce complexity, and boost generalization in ECG classification tasks. Another notable example is the multi-path and multi-channel model introduced by Ayano et al. [26], which integrates 1D-CNN, BiLSTM, and attention mechanisms to enhance feature extraction. Similarly, Dissanayake et al. [27] proposed the DConv-LSTM-Net, which combines dilated convolutions with LSTM layers to effectively analyze short-term ECG signals. This architecture not only improves interpretability and performance but also reduces computational complexity compared to conventional models. More recently, Zhang et al. [28] proposed a hybrid CNN-Transformer architecture for ECG signal analysis, demonstrating improved accuracy and robustness compared to conventional methods

Traditional models like RNNs and 1D-CNNs rely on training from scratch, demanding large labeled datasets. In medical fields, acquiring diverse, high-quality annotations is costly and time-consuming, posing a major challenge. Moreover, these models mainly capture temporal features and may overlook complex waveform shapes or spatial patterns across multiple leads, which are important for accurate heart diagnosis. To overcome these challenges, recent studies used pretrained deep networks trained on large, heterogeneous datasets. These models extract high-level features efficiently, speed up convergence, and reduce manual tuning, especially useful when training data is limited, imbalanced, or scarce. Numerous studies [10–12,29–35] have investigated the advantages of these models, and their results have shown that integrating pretrained networks with the transformation of ECG signals into two-dimensional images leads to improved accuracy and stability in classification models. Other studies have employed domain transformation methods such as CWT [29,12,10], Fast Fourier Transform (FFT) [29], and ST [12] to provide a more informative time-frequency representation of ECG signals. These methods have enabled more effective extraction of latent patterns, particularly when applied in models such as ResNet-22, AlexNet, SqueezeNet [30], Xception, VGG-19, DenseNet-121, and ResNetV2 [12], resulting in improved classification accuracy. In the study by Moran et al. [10], the AlexNet architecture was modified by adjusting its final layers, including the addition of a Softmax layer and a custom Fully Connected (FC) layer tailored for ECG, and it demonstrated superior performance in classifying cardiac signals.

Furthermore, in a study by Ullah et al. [31], investigated Premature Ventricular Contractions (PVCs) detection using a pretrained ResNet-18. They preprocessed MIT-BIH and INCART datasets into 2D images and tested the model in four scenarios. Results showed that combining pretrained models with visual ECG representations notably improves arrhythmia detection accuracy and reliability. In the study conducted by Mathunjwa et al. [11], the Recurrence Plot (RP) method was employed to analyze recurring patterns within ECG data, and deep learning models such as VGG-16, VGG-19, and AlexNet were applied for the classification of cardiac abnormalities. Similarly, Li et al. [32] utilized Empirical Mode Decomposition (EMD) to extract intrinsic mode functions embedded in ECG signals. This method, which adaptively decomposes signal components in accordance with their local variations, was integrated with the VGG-16 network and led to improved accuracy in detecting cardiac abnormalities. Moreover, the study by Jothiaruna and Leema [33] proposed a novel approach for identifying cardiac disorders based on twelve-lead digital ECG images. In this work, models such as SSDMNV2-FPN, MobileNetV2 were employed for ECG lead localization and enhancement of diagnostic accuracy.

Although Pretrained neural networks effectively extract features and improve classification but are often too large and computationally heavy for resource-limited devices like wearables. Qin et al. [14] proposed MVKT (Multi-View Knowledge Transfer), where a teacher model trained on twelve-lead ECG transfers knowledge to a single-lead student model, achieving higher accuracy than traditional distillation methods and enabling lightweight models for practical use. Following this, Qiang et al. [15] developed Multi-Task Multi-View KD Framework (MT-MV-KDF), a framework combining multi-task, multi-view learning, and KD for myocardial infarction detection and localization, improving accuracy, reducing complexity, and enhancing generalizability. Yang et al. [16] tackled multi-label ECG classification using multi-view and multi-scale networks, dividing signals into six lead groups and employing KD to create faster, smaller student models. Zhang et al. [36] also made a significant contribution to the optimization of deep learning models by introducing the Dual Distillation Double Gains)DDDG) method. This approach, which combines generative self-supervised learning with bidirectional KD, has led to a substantial reduction in model complexity while improving computational efficiency.

Collectively, these studies highlight that integrating advanced knowledge distillation techniques with deep learning can yield models that not only achieve high accuracy but also remain computationally efficient, making them well-suited for portable and resource-constrained medical devices, such as wearable ECG monitors. Nevertheless, despite these notable advances, several challenges persist. Most existing KD-based ECG classification frameworks rely on single-teacher or single-level knowledge transfer, which limits the diversity and richness of the information conveyed to the student model. Additionally, many methods focus primarily on response-level distillation, often neglecting intermediate feature representations that capture critical morphological and temporal patterns in ECG signals. Importantly, the use of twelve-lead ECGs provides a comprehensive view of cardiac activity from various angles, offering rich and complementary information that substantially enhances diagnostic accuracy. Taken together, these considerations underscore the need for a more flexible, hierarchical knowledge distillation framework capable of effectively balancing accuracy, interpretability, and computational efficiency.

Building upon the reviewed literature, this study proposes a HMT-KD framework aimed at improving both the accuracy and efficiency of ECG-based heart disease detection. Pretrained convolutional neural networks

have demonstrated strong capabilities in extracting deep and generalizable features from physiological signals, including ECGs. In this framework, ResNet-18 integrated with a SENet module was chosen as one of the teacher networks. Its residual connections, coupled with channel-wise attention, facilitate the extraction of rich morphological and amplitude-related features from ECG waveforms. Additionally, MobileNetV2, leveraging Depthwise Separable Convolutions, provides efficient and expressive feature extraction from ECG images. The integration of these two pretrained architectures within the proposed framework provides effective and diverse insights into ECG data. Owing to their strong representational capabilities, both networks demonstrate high potential in extracting valuable ECG features that contribute to accurate and robust classification of cardiac abnormalities. This architectural diversity underpins the hierarchical KD process, enabling the transfer of richer and more varied information to the student model.

Inspired by the architectures of the two primary teacher networks, a TA model was specifically designed to better capture and interpret the knowledge embedded in the teachers. The TA is initially trained through response-based distillation from the main teachers and subsequently facilitates knowledge transfer to the lightweight student model via both response-level and feature-level pathways. This multi-level and multi-source knowledge transfer occurs through the main teacher networks and the TA, effectively enriching the student model's understanding. The student network, which has a similar structure to the TA but with a more compact architecture, is thus able to achieve performance that surpasses the individual teacher networks. The lightweight student model thus demonstrates high accuracy, robustness, and generalizability, making it particularly suitable for deployment in wearable, portable, and smart healthcare devices. By combining diverse and effective information from multiple teachers with a carefully designed distillation strategy, the framework provides an efficient, real-time, and clinically relevant solution for automated cardiovascular monitoring.

3. Proposed method

This section introduces the overall framework of the proposed method and highlights its key technical components. The methodology is structured into five main steps: (1) dataset preparation, (2) implementation and training of the first teacher model, (3) implementation and training of the second teacher model, (4) development and training of the TA model, and (5) application of the HMT-KD framework, which leverages multi-path knowledge transfer to effectively guide the student model. Collectively, these stages are designed to improve diagnostic accuracy and computational efficiency, making the proposed framework well-suited for both real-time clinical applications and wearable ECG monitoring systems. As illustrated in Fig. 1, the proposed framework consists of five steps. In the first step, raw ECG signals are converted into two-dimensional image representations, following widely adopted standard ECG image formats. These images then undergo a pre-processing procedure.

In recent years, very powerful deep models such as ResNet and MobileNet have been presented for analyzing image data. These models, which have been trained with a large amount of image data, have shown

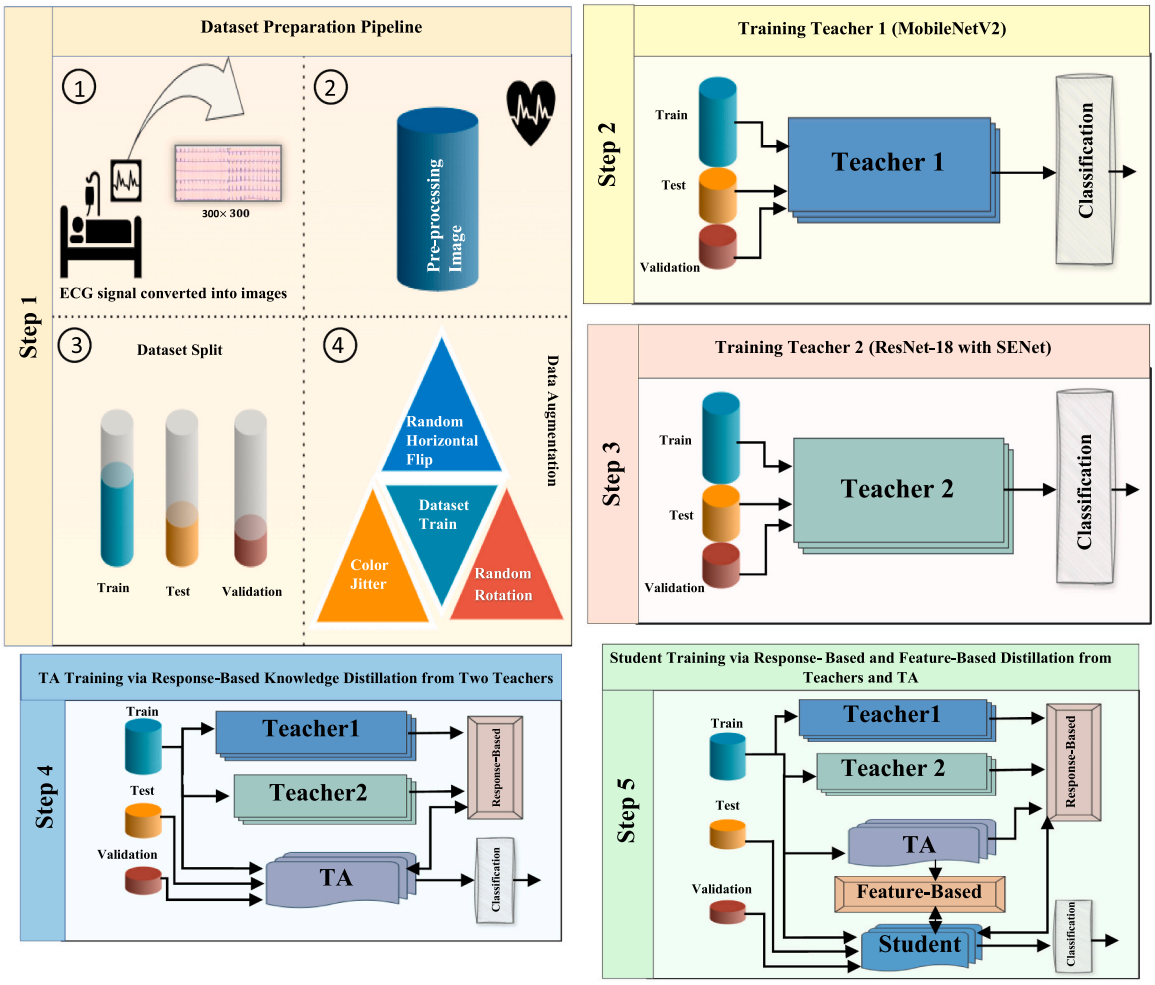


Fig. 1. An overview of five steps pipeline of the proposed Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework.

their efficiency in many applications. To use these pre-trained models and use the transfer learning technique, the type of input data must be consistent with the type of input of these models, which is images. By converting ECG signals to images, in addition to the ability to apply pre-trained models to the ECG images, data augmentation techniques such as rotation, flipping, shifting, and adding noise can also be used to increase the data required for training the models. Additionally, images with dimensions of 300 × 300 will provide deep models with much more data volume and features for disease diagnosis than ECG signals with dimensions of 12 × 5000. On the other hand, when the image of ECG signals is provided as input to the proposed model, it can be used both for clinical applications that have ECG signal data (by converting the signal to an image) and for home applications that have a printed version of the ECG (by scanning the printed version). So, the input data is converted into 300 × 300 pixels images.

Next, the data are split into training, validation, and test sets. To enhance generalization and reduce overfitting, data augmentation techniques are applied exclusively to the training set. In the second step, the first primary teacher model,MobileNetV2, is trained using the pre-processed ECG images. In the third step, the second primary teacher model, ResNet-18 enhanced with SENet, is trained on the same dataset. Together, these two teacher models form a strong foundation for effective knowledge transfer. In the fourth step, a TA network is introduced and trained through response-based knowledge distillation from both teachers. This process allows the TA to acquire rich feature representations from both teachers while improving its ability to guide the student model effectively. Finally, in the fifth step, the lightweight student network is trained through a dual-path knowledge distillation strategy. In one path, the student learns directly from the outputs of both teachers and the TA, while in the other path, it benefits from feature-based distillation drawn from the intermediate layers of the TA. Through the combination of these two forms of guidance, the student network achieves efficient knowledge transfer, high diagnostic accuracy, and compactness. This multi-step, multi-path design enables the proposed framework to achieve reliable performance in real-time applications, even under constrained computational resources.

3.1. Data augmentation and pre-processing

Key augmentations included horizontal flipping with a probability of 0.7 to increase robustness to spatial orientation. Additionally, color jittering was employed to simulate real-world variations, adjusting brightness (±0.2), contrast (±0.3), saturation (±0.2), and hue (±0.2). Additionally, a random rotation with a very limited range was applied. The intent of using such a small range was to simulate minor variations in image acquisition angles without distorting the delicate structural details present in the original data.These transformations enriched data diversity and improved the model's resilience to visual variability. After applying the aforementioned transformations, the images are converted into tensors, and their pixel values are normalized from the original range of [0,255] to [0,1]. To ensure consistency across the input data, Z-score standardization is then applied. In this process, the mean and standard deviation of each color channel $c \in \{R, G, B\}$ are calculated as shown in Eq. (1):

$$\mu_c = \frac{1}{N}\sum_{i=1}^{N} mean(X_i^c) \tag{1}$$

where N denotes the total number of training images and c refers to the selected color channel. Subsequently, the mean standard deviation for each channel $c = \{R, G, B\}$ is computed in a similar manner, as described below:

$$\sigma_c = \frac{1}{N}\sum_{i=1}^{N} std(X_i^c) \tag{2}$$

Finally, standardization of each pixel (x,y) in the image i is performed using the previously computed values, as expressed in the following equation:

$$Z_i^c(x,y) = \frac{X_i^c(x,y) - \mu_c}{\sigma_c}, \quad for\ c \in \{R, G, B\} \tag{3}$$

This preprocessing step helps the models receive well-balanced input data by minimizing the effect of extreme pixel values. As a result, it promotes more stable and efficient training, which ultimately improves the model's overall performance.

3.2. MobileNetV2 architecture

As illustrated in Fig. 2, MobileNetV2 [37] was developed with efficiency in mind, particularly for devices with limited computational resources. The architecture employs a series of inverted residual blocks with linear bottlenecks, allowing it to minimize computational cost without sacrificing performance. The model begins with a conventional convolutional layer, followed by multiple inverted residual blocks that make up the network's deeper structure. Each of these blocks includes three stages: an expansion layer, a depthwise convolution, and a projection layer. At the classification head of the network, a fully connected layer is introduced along with a dropout layer set to 0.2. During training, around 20% of the neurons in this layer are randomly deactivated. This helps prevent overfitting by reducing the model's reliance on specific features and encourages better generalization on new, unseen data.

To enhance the model's accuracy, specific layers of the network were selectively frozen, while others remained trainable. Freezing refers to keeping the weights of certain layers unchanged during training, thereby reducing computational costs and accelerating convergence. Moreover, by preserving critical feature representations, this strategy minimizes the risk of overfitting. When combined with dropout, it further improves generalization, resulting in more stable and reliable performance on unseen data. Table 1 provides a detailed breakdown of the frozen and trainable layers.

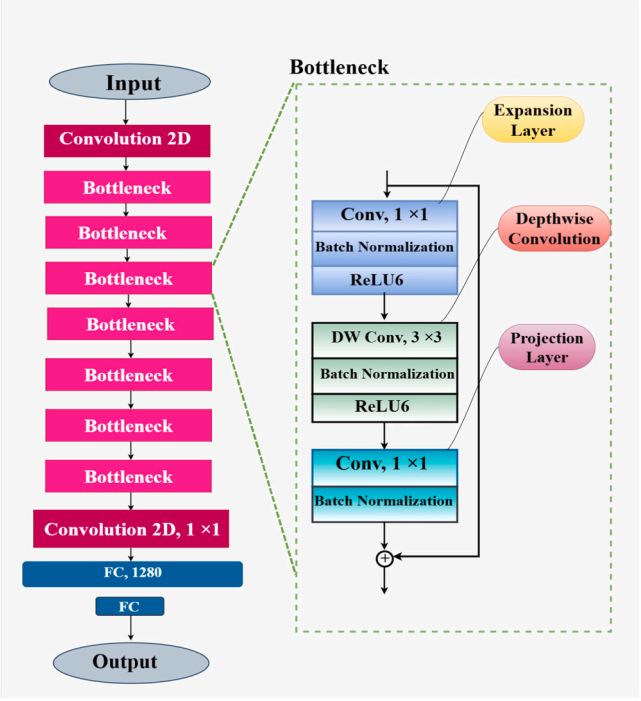


Fig. 2. The architecture of MobileNetV2.

Table 1
List of all layers in the MobileNetV2 architecture.

Training Status	Number of Filters	Layer
Trainable	32	Conv2d
Trainable	16	Bottleneck
Trainable	24 (×2)	Bottleneck
Trainable	32 (×3)	Bottleneck
Trainable	64 (×4)	Bottleneck
Trainable	96 (×2)	Bottleneck
Frozen	96	Bottleneck
Frozen	160 (×3)	Bottleneck
Frozen	320	Bottleneck
Frozen	1280	Conv2d 1 × 1
Trainable	Number of Classes	FC

3.3. ResNet-18 enhanced with SENet

In this study, ResNet-18 was integrated with the Squeeze-and-Excitation Network (SENet) module to enhance the model's ability to emphasize critical features in ECG images [38]. As depicted in Fig. 3, SENet modules were inserted after the third and fourth convolutional layers. These modules adaptively recalibrate channel-wise feature responses, allowing the network to highlight informative features while suppressing less relevant ones. The integration process involves combining the outputs of convolutional layers via skip connections, followed by the application of the SENet module to the aggregated features. This design significantly improves the model's capacity for feature discrimination and results in more accurate and robust ECG classification performance.

The implementation of the SENet module, illustrated in Fig. 3, consists of two primary stages: the squeeze phase and the excitation phase. In this process, the input ECG image, denoted as $X \in R^{H\times W\times C}$, is first passed through the ResNet-18 network. After processing through the residual blocks, a set of convolutional filters $P = [p_1, p_2, \ldots, p_{C'}]$ is applied, resulting in the extraction of feature maps represented as $U \in R^{H'\times W'\times C'}$.

In the squeeze phase, global average pooling is applied to compress the information within each channel. Specifically, each feature channel in the extracted feature map $U = [u_1, u_2, \ldots, u_{C'}] \in R^{H'\times W'\times C'}$ is processed to compute its average value. Here, H', W' and C' represent the height, width, and number of feature channels, respectively. The output of this stage is calculated as shown in Eq. (4).

$$z_k = \frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} u_k \ , \ for\ k = 1, \quad 2, \quad \ldots, \quad C' \tag{4}$$

The compressed feature vector $z \in R^{C'}$, which represents the aggregated information from each channel, is passed through a small neural network to learn the importance of each channel. This network consists of two fully connected layers. First, the initial fully connected layer $FC_1 \in \mathscr{R}^{\frac{C'}{r}\times r}$ is applied to the input vector, and the resulting output is processed by a ReLU activation function, denoted as δ. The output is then passed through a second fully connected layer $FC_2 \in \mathscr{R}^{C'\times\frac{C'}{r}}$, followed by a sigmoid activation function (σ). In this process, the reduction ratio r is set to 16. These operations are summarized in Eq. (5).

$$s = F_{ex}(z, FC) = \sigma(g(z, FC))\sigma(FC_2\delta(FC_1(z))) \tag{5}$$

The vector s, representing the attention weights for each channel, is multiplied element-wise with the feature maps extracted from the convolutional layer. As a result, the weighted feature vector $\widetilde{X} = [\widetilde{x}_1, \widetilde{x}_2, \ldots, \widetilde{x}_{C'}]$, is obtained. This process is defined by the following equation:

$$\widetilde{x}_k = \quad F_{scale}(u_k, s_k) = \quad u_k \bullet s_k \quad , \quad for\ k = 1, \quad 2, \quad \ldots, \quad C' \tag{6}$$

Consequently, channels assigned higher attention weights are amplified, while those with lower weights contribute less prominently. This attention mechanism allows the model to focus selectively on the most informative features within each ECG image, thereby improving both the accuracy and effectiveness of feature extraction. To enhance generalization and mitigate overfitting, the final classification stage incorporates a fully connected layer followed by a dropout layer with a rate of 0.5. This regularization technique strengthens the model's robustness and supports more reliable performance on previously unseen data.

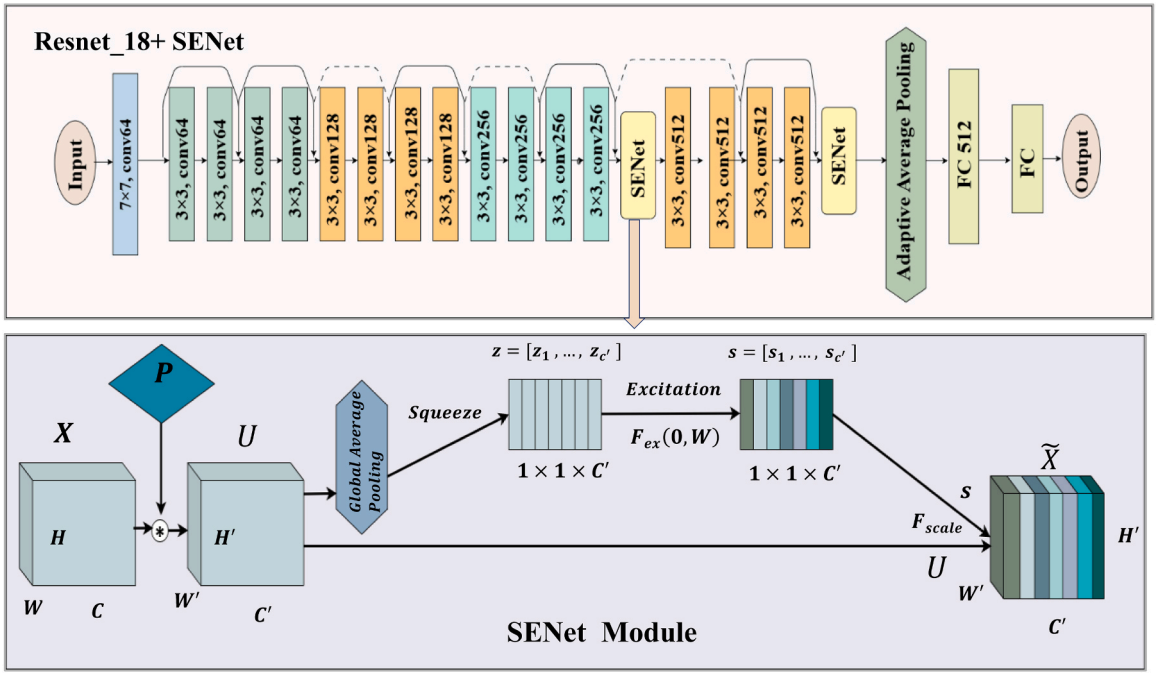


Fig. 3. The integration of the ResNet-18 model and the SENet module.

3.4. The structure of the proposed student and teacher assistant networks

The Student Network is designed to strike a balance between performance and efficiency. Drawing inspiration from both Teacher models, it employs key components such as DSC, RB, and the CBAM attention mechanism to capture meaningful patterns while maintaining low computational cost. Fig. 4 illustrates the Residual Block and CBAM modules, which play a central role in the network's architecture. Each component contributes to the model's ability to learn rich feature representations, even in resource-constrained environments.

In the early stage of the model, the input image passes through a Depthwise Separable Convolution layer, which includes two distinct steps. First, depthwise convolution applies a separate filter to each input channel individually. Then, a pointwise convolution (a 1 ×1 convolution) is used to integrate information across channels. This efficient design reduces both the parameter count and computational load, without compromising the quality of feature extraction. The resulting features are then standardized using Batch Normalization, followed by a MaxPooling layer that decreases the spatial resolution, retaining the most critical features while further optimizing computational performance.

The extracted features are subsequently passed through a sequence of residual blocks, each consisting of two Depthwise Separable Convolution layers, with each layer followed by Batch Normalization. These blocks incorporate shortcut (or skip) connections that enable the original input to bypass the primary convolutional path. This architectural design helps mitigate the vanishing gradient problem and facilitates the training of deeper networks, thereby improving the model's overall accuracy and representation learning capabilities. To enhance the model's ability to focus on the most relevant regions of the image, the CBAM [39] is also employed. In this study, a lightweight variant of CBAM comprising only the Channel Attention module is utilized to minimize model complexity and increase the efficiency of ECG image processing. By excluding the Spatial Attention component, both the parameter count and computational cost are substantially reduced, which is particularly advantageous in ECG-related applications where resource efficiency is critical.

In the proposed attention module, feature descriptors for each channel are initially extracted through two parallel pathways: Adaptive Average Pooling and Adaptive Max Pooling. While the average pooling pathway provides a global and smoothed representation of the feature distribution, the max pooling pathway focuses on emphasizing salient regions and preserving sharp edges. The combination of these perspectives generates a more informative attention map, enabling the model to effectively concentrate on the most significant features and thereby improving feature extraction performance. Subsequently, each channel descriptor is fed into a lightweight sub-module consisting of two consecutive 1 × 1 convolutional layers, which function equivalently to a multilayer perceptron (MLP). The first convolutional layer reduces the channel dimension by a factor of 16, serving as a compression step to optimize computational efficiency. The output is then activated by a ReLU function, followed by restoration of the original dimensionality through the second convolutional layer. This design not only reduces the number of parameters but also facilitates the learning of complex and non-linear feature interactions, which is crucial for enhancing the attention mechanism. Finally, outputs from both pooling pathways are aggregated and passed through a sigmoid activation function to generate the channel attention map $M_c(x)$.

This attention map is applied element-wise to the original input x, allowing the network to selectively emphasize informative features while suppressing less relevant ones. As a result, the model achieves improved precision and focus in extracting key representations from ECG images. The corresponding formula can be expressed as follows:

$$M_c(x) = sigmoid((conv2(ReLU(conv1(AvgPool(x))))) + (conv2(ReLU(conv1(MaxPool(x))))) \quad (7)$$

$$x' = M_c(x) \times x \quad (8)$$

In the final stage of the model, following Adaptive Average Pooling and the transformation of features into a fixed-size vector, a sequence of three FC layers is applied. Initially, the vector is expanded from 64 to 128 dimensions. This is followed by Batch Normalization, a ReLU activation function, and a Dropout layer with a rate of 0.5 measures that enhance training stability and reduce the risk of overfitting. The second

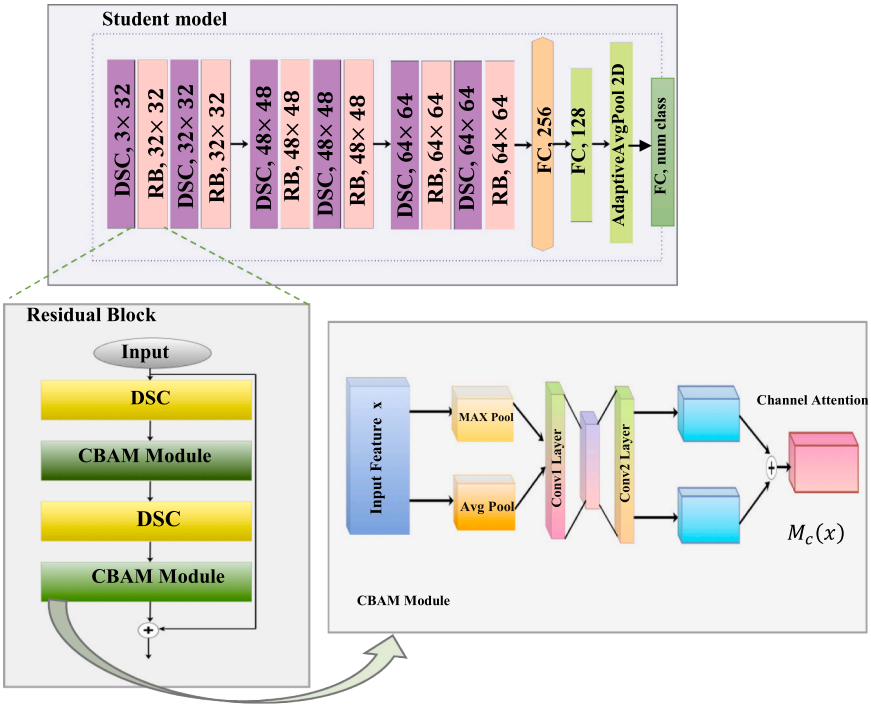


Fig. 4. The Residual Block and CBAM modules used in the proposed Student model.

FC layer adopts a similar structure, further increasing the dimensionality to 256. Finally, the last Fully Connected layer projects the learned features into an output space corresponding to the number of target classes, enabling the final classification. The proposed model, whose architecture is detailed in Table 2, integrates deep learning techniques with attention mechanisms to provide a lightweight, optimized, and efficient framework for ECG analysis. This design demonstrates considerable potential for medical diagnosis and vital signal interpretation, offering a robust tool for accurate and effective assessment. Overall, the student model combines modern techniques DSC to lower computation, residual connections for deep learning, and a lightweight CBAM attention module to highlight key features. Its design balances simplicity and efficiency while maintaining high accuracy in ECG feature extraction, making it well-suited for real-time systems with limited resources.

The TA model adopts a deeper and more complex architecture than the student model, leveraging multiple stages of depthwise separable convolutions, residual blocks, and CBAM modules to improve feature extraction capabilities. By combining global average and max pooling, it captures richer representations, and with three fully connected layers, it effectively maps features to the final classification. Despite its increased depth, the model maintains efficiency through separable convolutions. The detailed architecture of the model is presented in Table 3.

3.5. Knowledge distillation

This section introduces an innovative HMT-KD strategy designed specifically for ECG image-based analysis. The distillation process is conducted in two stages to maximize knowledge transfer efficiency.

3.5.1. Teacher assistant training

As illustrated in Fig. 5, in the first stage of the proposed framework, two high-performance teacher models MobileNetV2 and ResNet-18 with SENet are employed to train a TA network. To combine their knowledge effectively, a specificity-based weighting strategy is applied. For each input sample x_i, the teachers produce output logits M_{ij}^1 and M_{ij}^2 for class j, which are weighted by their respective specificity θ_1, θ_2:

$$Z_{ij}^{total} = \theta_1 \times M_{ij}^1 \quad + \theta_2 \times M_{ij}^2 \tag{9}$$

The aggregated logits (Z_{ij}^{total}) are then passed through a temperature-scaled Softmax to obtain softened probabilities [40,41]:

$$p_{ij}^{total(TA)} = \frac{\exp(Z_{ij}^{total} \Big/ T)}{\sum_{i=1}^{n}\sum_{j=1}^{k}\exp(Z_{ij}^{total} \Big/ T)} \tag{10}$$

In Eq. (10), $p_{ij}^{total(TA)}$ represents the temperature-scaled soft predictions from the teachers, providing richer supervisory signals than hard labels. A higher temperature T produces a smoother probability distribution, facilitating better knowledge transfer. The TA network outputs logits M_{ij}^{TA}, which are converted into its own softened predictions $q^{ij(TA)}$ using the same temperature-scaled Softmax:

$$q^{ij(TA)} = \frac{\exp(M_{ij}^{TA} \Big/ T)}{\sum_{i=1}^{n}\sum_{j=1}^{k}\exp(M_{ij}^{TA} \Big/ T)} \tag{11}$$

The second component, y_{ij}^{TA}, represents the hard label for the $i-th$ sample in class j. The hard-label loss for the TA model is defined as:

Table 2
The architecture and layer details of the proposed student model.

Stage	Layer Type	Input Channels	Output Channels	Description
1	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	3	32	Initial depthwise separable convolution layer
2	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	32	32	Second separable convolution layer
3	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	32	48	Third separable convolution layer
4	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	48	48	Fourth separable convolution layer
5	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	48	64	Fifth separable convolution layer
6	DepthwiseSeparableConv + BatchNorm + ReLU + MaxPool	64	64	Sixth separable convolution layer
7	AdaptiveAvgPool2d	64	64	Global average pooling
8	Fully Connected + BatchNorm + ReLU + Dropout	64	128	First fully connected layer
9	Fully Connected + BatchNorm + ReLU + Dropout	128	256	Second fully connected layer
10	Fully Connected (Output Layer)	256	Number of class	Final classification layer

Table 3
The architecture and layer details of the proposed Teacher Assistant model.

Stage	Layer Type	Input Channels	Output Channels	Description
1	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	3	32	Initial convolutional stage with residual block
2	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	32	32	Second convolutional stage with residual block
3	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	32	64	Third convolutional stage with increased channels
4	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	64	64	Fourth convolutional stage with residual block
5	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	64	128	Fifth convolutional stage with increased channels
6	DepthwiseSeparableConv + BatchNorm + ReLU + ResidualBlock + MaxPool	128	128	Sixth convolutional stage with residual block
7	AdaptiveAvgPool2d + AdaptiveMaxPool2d (concatenated)	256 (128 ×2)	128	Global average and max pooling, concatenated
8	Fully Connected + BatchNorm + ReLU + Dropout	128	128	First fully connected layer
9	Fully Connected + BatchNorm + ReLU + Dropout	128	64	Second fully connected layer
10	Fully Connected (Output Layer)	64	Number of class	Final classification layer

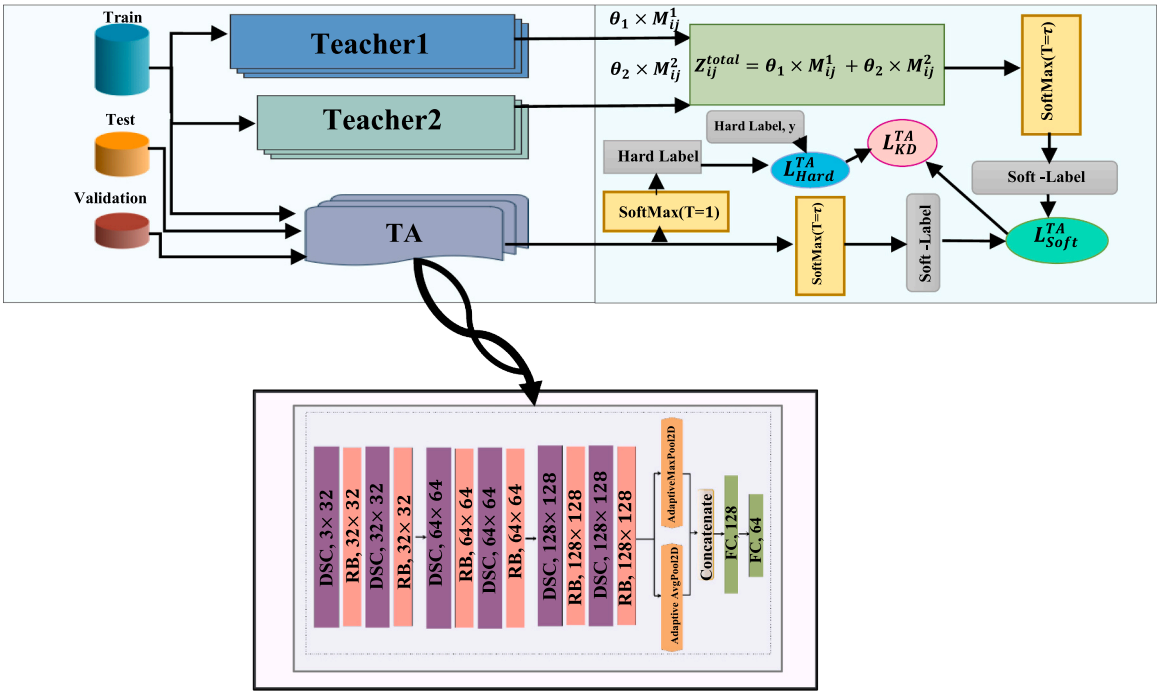


Fig. 5. The Teacher Assistant (TA) training process, for effectively transferring knowledge from the TA and two primary teachers to the student network.

$$L_{Hard}^{TA} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} y_{ij}^{TA} \log q^{ij(TA)} \tag{12}$$

To align the TA's soft predictions $q^{ij(TA)}$ with the aggregated soft targets $p_{ij}^{total(TA)}$ from both teachers, the Kullback–Leibler (KL) divergence is used:

$$L_{Soft}^{TA} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} p_{ij}^{total(TA)} \log \frac{p_{ij}^{total(TA)}}{q^{ij(TA)}} \tag{13}$$

The final TA training objective combines both soft and hard losses:

$$L_{KD}^{TA} = \beta L_{Soft}^{TA} + (1-\beta) L_{Hard}^{TA} \tag{14}$$

As a result, the TA model, after being trained by the two primary teacher networks and enriched with their combined knowledge, is strengthened in both its feature representation capacity and its ability to extract complex patterns. This enhanced capability enables the TA to serve as an effective intermediary between the teachers and the student network, facilitating a smoother and more efficient knowledge transfer process in subsequent stages, particularly during the training of the student model.

3.5.2. Student training

In this stage, as illustrated in Fig. 6, the proposed approach for training a lightweight student network is introduced, built upon a hierarchical framework that leverages three models: two primary teachers and one TA. Unlike conventional methods that rely solely on the final outputs of teacher networks, this approach combines two distinct knowledge transfer pathways: response-based and feature-based distillation. In the response-based pathway, the final outputs (logits) of all three models (ResNet-18 enhanced with a SENet module, MobileNetV2, and the TA) are used to guide the training of the student model. A specificity-based weighting mechanism is applied to adjust the influence of each teacher according to its ability to minimize false positives on healthy samples. In the feature-based pathway, intermediate representations from the TA which has been previously strengthened through response-based distillation from the two main teachers are transferred to the student model. In this setting, the TA serves as a powerful intermediary, delivering deep and structured knowledge from its internal layers to the student. By jointly leveraging both high-level information (logits) and intermediate representations (feature maps), this hybrid learning strategy enhances classification accuracy, improves generalizability, and increases the robustness of the student model when faced with new input data.

Within this distillation framework, an input sample x_i is simultaneously processed by the pre-trained teacher models and the TA network. Their respective outputs, M_{ij}^1, M_{ij}^2, and M_{ij}^3 represent the predicted probability that the $i-th$ sample belongs to class $j-th$, as inferred by the teacher and TA models. Then, if θ_1, θ_2 and θ_3 represent the specificity values obtained for the teacher and TA models, respectively, the following steps should be taken:

$$M_{ij}^{total} = \theta_1 \times M_{ij}^1 + \theta_2 \times M_{ij}^2 + \theta_3 \times M_{ij}^3 \tag{15}$$

Ultimately, the final soft predictions of all models are computed after applying the Softmax function, as shown below:

$$p_{ij}^{total} = \frac{\exp(M_{ij}^{total}/T)}{\sum_{i=1}^{n}\sum_{j=1}^{k}\exp(M_{ij}^{total}/T)} \tag{16}$$

In Eq. (16), the value p_{ij}^{total} denotes a softened prediction obtained by applying a temperature hyperparameter T. Instead of relying solely on hard ground-truth labels, the student model is guided by these softened outputs from the teacher networks, which provide a richer and more informative supervisory signal. Increasing the temperature T smooths the predicted probability distribution, thereby enabling a more effective knowledge transfer from the teacher to the student network.

Moreover, the output of the student model consists of two components. The first component, denoted as M_{ij}^s, represents the output of the final fully connected layer in the student classifier for the $i-th$ sample belonging to the $j-th$ class. Based on this output, the softened prediction q^{ij} is computed using Eq. (17):

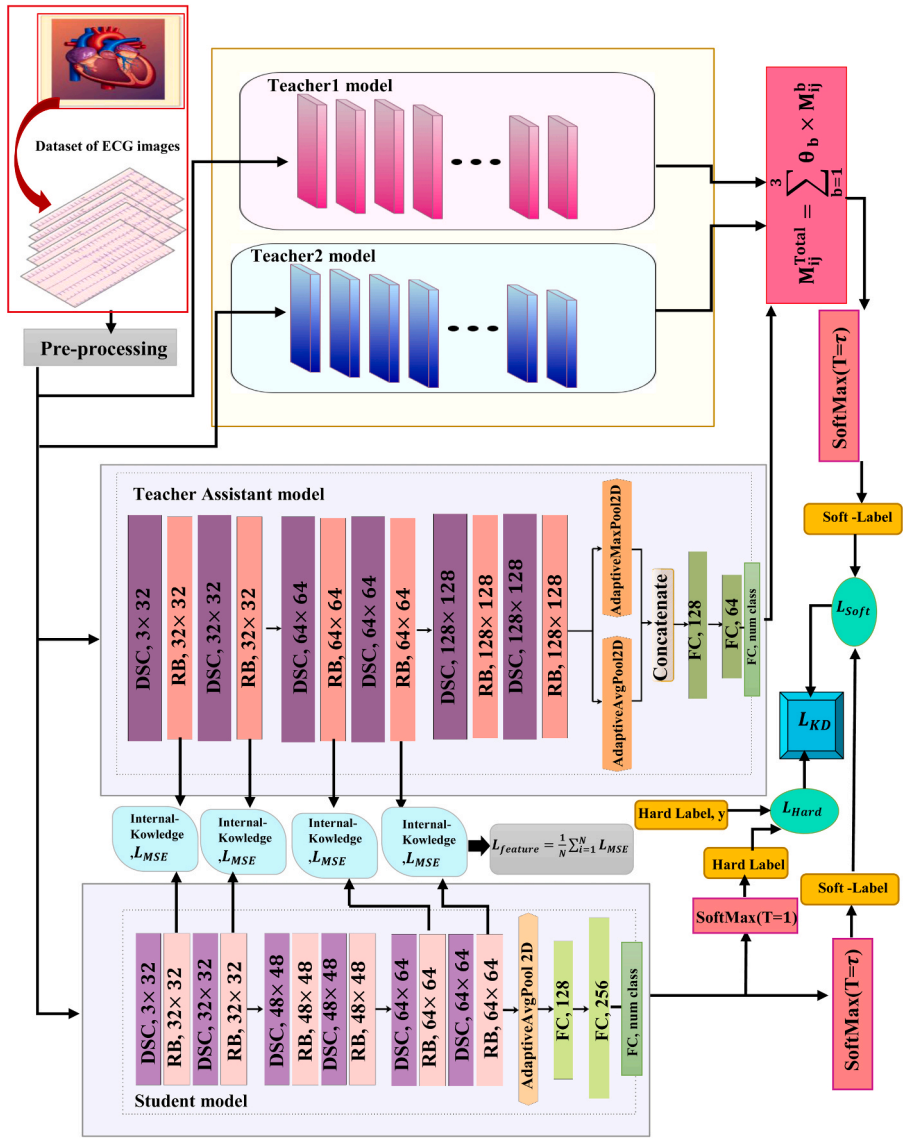


Fig. 6. A schematic overview of the proposed Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework, integrating response-based and feature-based pathways. The response-based KD exploits the outputs of ResNet-18 with SENet, MobileNetV2, and the Teacher Assistant, while the feature-based KD transfers intermediate representations from the TA to the student.

$$q^{ij} = \frac{\exp(M_{ij}^{s}/T)}{\sum_{i=1}^{n}\sum_{j=1}^{k}\exp(M_{ij}^{s}/T)} \tag{17}$$

The second component, denoted as y_{ij}, represents the hard label corresponding to the $i-th$ sample of class j. The loss function, which quantifies the discrepancy between the soft labels and the hard labels in the student model, is computed as follows:

$$L_{Hard} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} y_{ij}\log q^{ij} \tag{18}$$

To quantify the discrepancy between the soft label distributions generated by the teachers and student models, the KL Divergence is employed. This widely used metric measures the difference between two probability distributions and is a standard choice in knowledge distillation for assessing the divergence between model outputs. In this process, the KL Divergence determines how well the student model's output aligns with the outputs of the teacher models. A lower KL divergence value indicates greater similarity between the two distributions, reflecting a more effective knowledge transfer from the teacher to the student. The detailed formulation of this calculation is provided in Eq. (19).

$$L_{Soft} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} p_{ij}^{total}\log\frac{p_{ij}^{total}}{q^{ij}} \tag{19}$$

Here, n denotes the total number of samples, and k represents the number of classes in the dataset. Since the student model learns not only from hard labels but also from the soft predictions provided by the teacher models, a comprehensive loss function is formulated to guide the KD process. This loss function, detailed in Eq. (20), is essential for transferring knowledge from the teacher models to the student. By integrating information from both hard labels and soft targets, it optimizes the student's learning and enhances its ability to capture underlying data patterns.

$$L_{KD} = \beta L_{Soft} + (1-\beta)L_{Hard} \tag{20}$$

In this formulation, β is a crucial hyperparameter that balances the impact of soft teacher outputs and hard ground-truth labels in the loss

function. Proper tuning of β ensures the student model effectively leverages both sources of information, improving KD.

In addition to the response-based distillation pathway, the proposed framework incorporates a feature-based KD strategy to facilitate the transfer of more structured and fine-grained representations from the TA model to the student model [42]. In this approach, intermediate feature maps are extracted from selected layers of both the TA and student networks during the forward pass. Since the number of channels in selected layers is aligned between the TA and the student, the features can be directly compared. However, spatial resolution mismatches occurred in some layers. To address this, bilinear interpolation is employed to spatially align the feature maps, ensuring they are comparable and enabling precise computation of the distillation loss [43].

Once aligned, the Mean Squared Error (MSE) between each pair of corresponding feature maps is computed and used as the feature distillation loss. This process enables the student model to not only mimic the final outputs of the teachers and TA models but also reconstruct the internal behavior and intermediate representations of the TA model. As shown in Eq. (21), f_i^{TA} and f_i^s denote the features extracted and aligned from the $i-th$ selected layer of the TA and student models, respectively, and N represents the total number of layers involved in this distillation:

$$L_{feature} = \frac{1}{N}\sum_{i=1}^{N} ||f_i^s - f_i^{TA}|| \tag{21}$$

To simultaneously optimize the student model using both response-based and feature-based KD losses, denoted as L_{KD} and $L_{feature}$ respectively, a combined loss function is defined. In this formulation, γ is a tuning hyperparameter that balances the contribution of the two distillation pathways response-based and feature-based distillation. The final objective function is defined as follows:

$$L_{Total} = (1-\gamma)L_{KD} + \gamma L_{feature} \tag{22}$$

4. Experimental

To ensure fair and consistent evaluation, all teachers and student models were implemented using the PyTorch framework and executed in the Kaggle environment using a Tesla P100 GPU. The student model was trained for 65 epochs with an initial learning rate of 0.0004, a weight decay of 1×10^{-5}, and a batch size of 128, using the CrossEntropyLoss function. The TA model was trained for 50 epochs, the ResNet-18 with SENet for 45 epochs, and the MobileNetV2 for 30 epochs, while the other training settings and results were similar to those of the student model. To optimize the model parameters, the AdamW optimizer was adopted, which effectively decouples weight decay from the gradient update, making it a suitable choice for regularized deep learning tasks. To enhance training stability and convergence, a ReduceLROnPlateau learning rate scheduler was employed. This scheduler monitors the validation loss and decreases the learning rate by a factor of 0.85 if no improvement is observed over three consecutive epochs.

In the context of KD, meticulous tuning of hyperparameters is essential. A key hyperparameter is the temperature value T, which regulates the softness of the output probability distribution. Following the recommendation in [44], the temperature was set to 2. To comprehensively evaluate the performance of the proposed models in heart disease classification, a set of well-established and effective metrics was employed. In this study, Accuracy, Recall (Sensitivity), Precision, Specificity, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUROC) were used as model evaluation tools. These metrics are defined as follows:

$$Recall = \frac{TP}{TP + FN} \tag{23}$$

$$Precision = \frac{TP}{TP + Fp} \tag{24}$$

$$F_1 = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)} \tag{25}$$

$$Specificity = \frac{TN}{TN + FP} \tag{26}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{27}$$

$$AUROC = \int_0^1 Curve\ ROC \tag{28}$$

In this study, two widely recognized ECG datasets Chapman and PTB-XL were employed to thoroughly assess the effectiveness of the proposed models, as presented in Fig. 7. The Chapman dataset [45], developed through a collaboration among Chapman University, Shaoxing People's Hospital, and Ningbo First Hospital, comprises ECG recordings sampled at 500 Hz. Each recording contains a 10-second segment of a standard 12-lead ECG. The dataset is categorized into four rhythm classes: Sinus Bradycardia (SB), Normal Sinus Rhythm (SR), Atrial Fibrillation (AFIB), and General Supraventricular Tachycardia (GSVT). For evaluation purposes, the data were randomly divided into training (70%), testing (20%), and validation (10%) subsets.

The PTB-XL dataset [46], on the other hand, is considered one of the most extensive and high-quality resources available for ECG signal analysis. It includes 21,799 recordings from 18,869 patients, each representing a 10-second, 12-lead ECG segment sampled at 500 Hz. A major advantage of this dataset lies in its expert annotation by two cardiologists, offering 71 diagnostic labels categorized into diagnostic, form, and rhythm classes. In this work, we focused on the super-diagnostic classification, which aggregates these labels into five clinically meaningful categories: Normal (NORM), Myocardial Infarction (MI), ST/T Change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). This high-level categorization facilitates a more targeted and interpretable analysis of diverse ECG patterns. It should be noted that this study exclusively included recordings sampled at 100 Hz.

4.1. Results of the ablation study

To evaluate the contribution of each component in the proposed KD framework, an ablation study was conducted on the PTB-XL dataset. This analysis focused on the impact of two hyperparameters, β and γ, which control the influence of the response-based and feature-based distillation paths, respectively. As shown in Eq. 20, the parameter β regulates the balance between the soft labels generated by the teacher models and the hard ground-truth labels. To determine its optimal value, β was varied from 0.0 to 0.9, and the student model was trained under each setting. The performance results are illustrated in Fig. 8-A. When β was set to 0.7, the student model achieved its highest F1-score (74.25%), along with strong precision (79.33%) and AUROC (96.87%) values. These results indicate that placing greater emphasis on the soft outputs of teacher models can significantly enhance student performance. Therefore, $\beta = 0.7$ was selected for all subsequent experiments.

Similarly, the analysis revealed that $\gamma = 0.2$ resulted in the highest F1-score of 76.22%, reflecting an optimal balance between high-level supervision and structured knowledge transfer from intermediate representations. Under this setting, the model also achieved strong recall (74.33%) and precision (80.74%) values, outperforming other configurations. Furthermore, the AUROC peaked at 96.53%, demonstrating the model's robustness under this configuration. The obtained results presented in Fig. 8-B.

When γ was set to 0, the student model relied solely on response-based supervision from the teacher networks, resulting in suboptimal

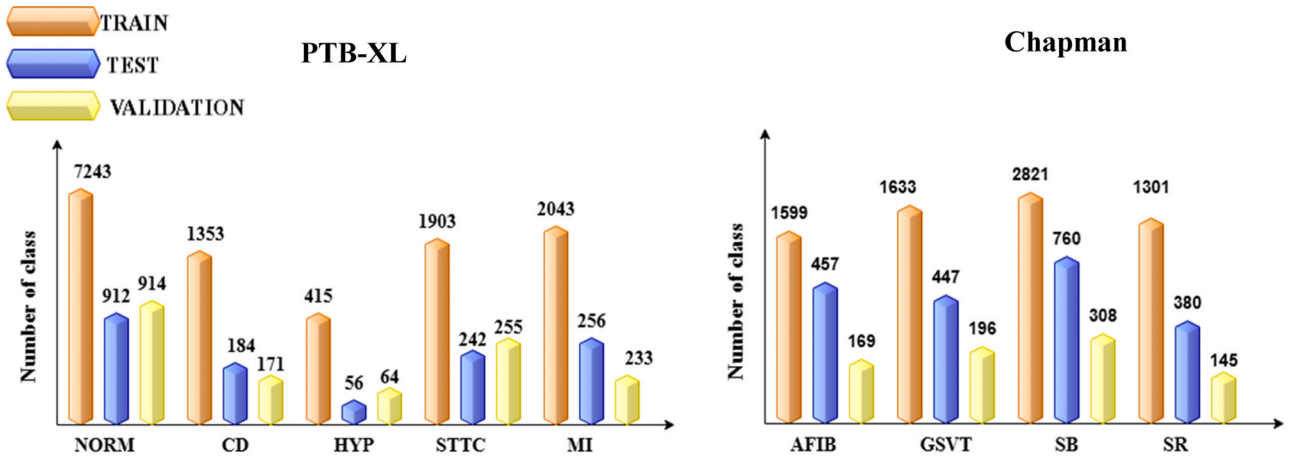


Fig. 7. The class distribution across training, validation, and test sets for the PTB-XL and Chapman datasets.

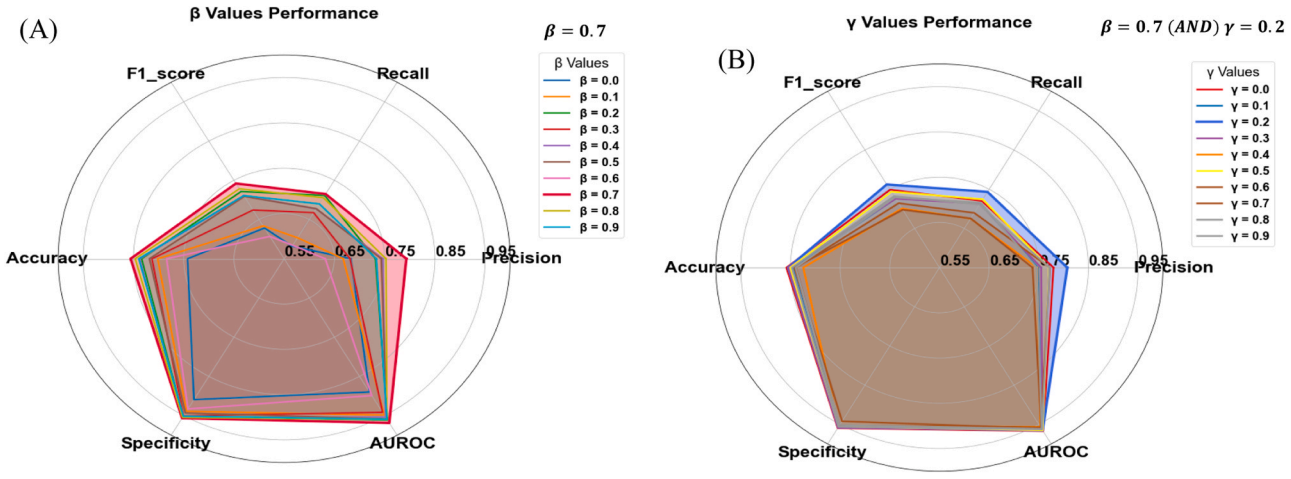


Fig. 8. The ablation analysis of the obtained results for evaluation of β and γ parameters. A) The model's performance under different values of β in the response-based distillation path. B) The effect of varying γ in the feature-based distillation path.

performance across most metrics. This outcome highlights the crucial role of feature-based knowledge transfer in enhancing generalization and robustness.

To further ensure the superiority of the Proposed Model over all other possible training configurations, the student network was evaluated with $\beta = 0.7$ and $\gamma = 0.2$ across various teacher and TA combinations on the PTB-XL dataset, with the results illustrated in Fig. 9.

As shown, the Proposed Model, leveraging both primary teachers

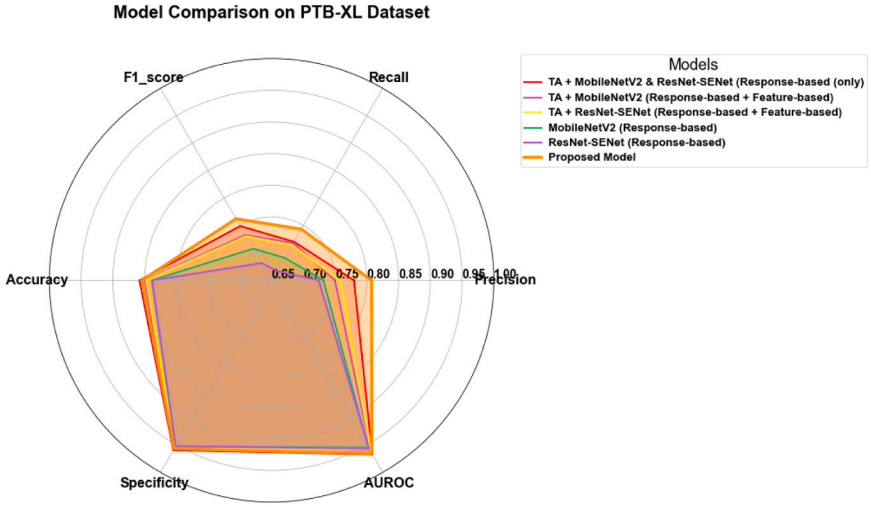


Fig. 9. The performance of the student model on PTB-XL with $\beta = 0.7$ and $\gamma = 0.2$ across different teacher and Teacher Assistant combinations.

(ResNet-18 with SENet and MobileNetV2) alongside the intermediate TA through a dual-path distillation strategy, consistently attains the highest scores across all evaluation metrics. Specifically, it achieves 80.74% Precision, 74.33% Recall, 76.22% F1-score, 85.45% Accuracy, 95.69% Specificity, and 96.53% AUROC, surpassing the student trained under any other considered combination of teachers and TA. These findings highlight that the HMT-KD framework effectively guides the student to acquire more comprehensive and robust representations. By integrating response-based and feature-based distillation, the proposed approach enables the lightweight student model to deliver high diagnostic accuracy while remaining computationally efficient and well-suited for real-time ECG monitoring applications.

Fig. 10 illustrates the performance of the TA network when trained using either a single primary teacher (MobileNetV2 or ResNet-SENet) or a combination of both, with β set to 0.7. As observed, training the TA with both primary teachers significantly enhances its overall performance across all metrics compared to using a single teacher. Specifically, the dual-teacher configuration achieves 79.33% Precision, 71.50% Recall, 74.20% F1-score, 85.58% Accuracy, 95.65% Specificity, and 96.87% AUROC, outperforming the single-teacher setups. These results highlight the benefit of integrating knowledge from multiple teachers when training the TA, as it enables the network to capture richer and more diverse feature representations. Consequently, the TA trained with two teachers serves as a more effective intermediary for guiding the student model, supporting improved diagnostic performance in downstream ECG classification tasks.

4.2. Results of PTB-XL dataset and Chapman dataset

To further validate the effectiveness of the proposed KD framework, the performance of the student model was compared against individual teacher models and the TA network using the PTB-XL dataset. The evaluation metrics included Precision, Recall, F1-score, Accuracy, Specificity, AUROC, and the number of models parameters, as presented in Table 4. Among the teacher models, ResNet-18 with SENet achieved the highest F1-score (75.65%) and AUROC (96.75%), while MobileNetV2 showed slightly lower performance but with fewer parameters. The TA network, which was trained using knowledge distilled from both teachers, achieved competitive results with an F1-score of 74.20% and the highest AUROC (96.87%), while being significantly more lightweight (194 K parameters). The student model trained without any KD exhibited significantly lower performance, with an F1-score of only 61.78% and an AUROC of 88.98%. In stark contrast, the proposed HMT-KD framework strategy enabled the student model to achieve an impressive F1-score of 76.22% and AUROC of 96.53%, despite having the same lightweight architecture (only 101 K parameters). This substantial improvement demonstrates that the proposed KD framework not only enhances predictive performance but also enables highly compact models to match or even outperform larger teacher networks in key clinical metrics.

To gain deeper insights into the model's behavior across different diagnostic categories, Table 5 presents a class-wise evaluation of the Student (With KD) model. The model demonstrated excellent performance in identifying normal (NORM) cases, achieving the highest F1-score (95.59%), recall (98.68%), and AUROC (99.50%).

For the CD class, the model reached a high precision of 90.97% and a specificity of 99.11%, indicating strong reliability in distinguishing pathological cases from healthy ones. While performance was relatively lower for the MI class (F1-score = 60.33%), the model still maintained reasonable AUROC (93.90%), suggesting moderate discriminative ability. The HYP and STTC classes showed balanced results with F1-scores of 68.63% and 76.68%, respectively. Overall, the model achieved a macro-averaged F1-score of 76.22% and AUROC of 74.33% across all five classes, indicating its robustness and generalization capability across diverse cardiac conditions.

The confusion matrix analysis Fig. 11 showed that the student model achieved excellent performance in distinguishing the Normal and STTC classes, with true positive rates of 98.68% and 89.67%, respectively. The model demonstrated relatively lower accuracy for the MI and HYP classes, with some misclassifications primarily into STTC and NORM. Specifically, 49.61% of MI samples were correctly identified, indicating room for further improvement in feature representation for these classes. Overall, these results highlight the model's strong capability in identifying common patterns, while suggesting opportunities for refinement in more challenging categories. Fig. 11 also shows that the training and validation loss values demonstrate effective learning by the model. The rapid and stable decrease in loss for both training and validation sets indicates that the model is learning well without significant overfitting. To more precisely evaluate the effectiveness and generalizability of the proposed KD framework, additional experiments were conducted on the Chapman dataset.

Table 6 presents a comparative evaluation of the primary teacher models, the TA network, and the student models with and without KD using several performance metrics including Precision, Recall, F1-score, Accuracy, Specificity, AUROC, and the number of parameters. Among the teacher networks, ResNet-18 with SENet achieved the highest F1-score (95.05%) and AUROC (99.44%), while MobileNetV2 exhibited comparable performance with fewer parameters, demonstrating its

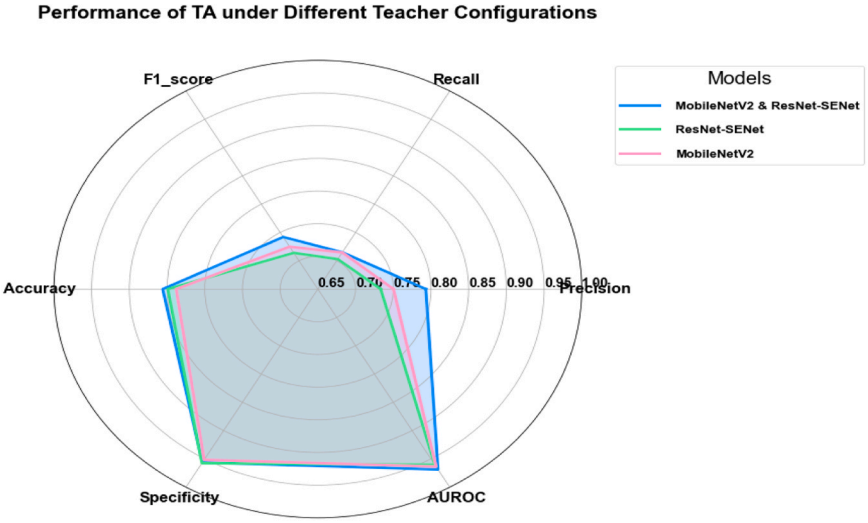


Fig. 10. The performance of single- or dual-teacher training of Teacher Assistant on PTB-XL.

Table 4
Performance comparison of teacher networks, Teacher Assistant (TA), and student model with and without knowledge distillation on the PTB-XL dataset.

Models	Precision	Recall	F1_score	Accuracy	Specificity	AUROC	Param
MobileNetV2 (Teacher)	0.7790	0.7090	0.7390	0.8539	0.9538	0.9619	2230277
Resnet_18 +SENet (Teacher)	0.8052	0.7267	0.7565	0.8527	0.9518	0.9675	11220853
TA (With KD)	0.7933	0.7150	0.7420	0.8558	0.9565	0.9687	194880
Student (NO KD)	0.6829	0.5819	0.6178	0.7424	0.9092	0.8898	101600
Student (With KD)	0.8074	0.7433	0.7622	0.8545	0.9569	0.9653	101600

Table 5
The performance of Student (With KD) Model on different class of PTB-XL Dataset.

Class	Precision	Recall	F1_score	Specificity	AUROC
CD	0.9097	0.7120	0.7988	0.9911	0.9679
HYP	0.7609	0.6250	0.6863	0.9931	0.9590
MI	0.7697	0.4961	0.6033	0.9727	0.9390
NORM	0.9269	0.9868	0.9559	0.9038	0.9950
STTC	0.6698	0.8967	0.7668	0.9240	0.9658
Total	0.8074	0.7433	0.7622	0.9569	0.9653

efficiency. The TA network, trained via KD from both teachers, attained an F1-score of 92.80% and AUROC of 98.80%, showing a strong balance between performance and lightweight design. The student model without KD performed reasonably well (F1-score = 91.10%, AUROC = 98.69%) despite its compact architecture. However, once trained using the proposed multi-teacher KD strategy, the student model with KD achieved significant performance gains across all metrics (F1-score = 93.74%, AUROC = 98.86%), approaching the effectiveness of larger teacher models while maintaining a drastically reduced parameter count (only ~101 K parameters). These findings highlight the capability of the proposed framework to produce compact yet highly accurate models suitable for deployment in resource-constrained environments.

The class-wise evaluation on the Chapman dataset, as presented in Table 7, highlights the strong discriminative ability of the student model trained with KD. Specifically, the model demonstrated outstanding performance on the SB and SR classes, achieving F1-scores of 98.56% and 94.71%, respectively, along with remarkably high AUROC values (99.84% and 98.54%). The results for AFIB and GSTV also remained competitive, with F1-scores exceeding 89%, indicating the model's ability to handle more complex and irregular rhythms effectively. Notably, the specificity across all classes remained above 96%, suggesting a low false positive rate and reliable classification. These



Fig. 11. Confusion matrix for the student model across five ECG classes on the PTB-XL dataset and training and validation loss and accuracy curves for the student network with KD.

Table 6
Performance comparison of teacher, Teacher Assistant (TA), and student models on the Chapman dataset.

Models	Precision	Recall	F1_score	Accuracy	Specificity	AUROC	Param
MobileNetV2 (Teacher)	0.9479	0.9478	0.9477	0.9535	0.9849	0.9952	2230277
Resnet_18 +SENet (Teacher)	0.9509	0.9503	0.9505	0.9502	0.9851	0.9944	11220853
TA (With KD)	0.9296	0.9269	0.9280	0.9359	0.9878	0.9880	194815
Student (NO KD)	0.9164	0.9086	0.9110	0.9207	0.9733	0.9869	101.343
Student (With KD)	0.9399	0.9356	0.9374	0.9442	0.9816	0.9886	101.343

Table 7
Class-wise Evaluation Results of the Student Model (With KD) on the Chapman Dataset.

Class	Precision	recall	F1_score	Specificity	AUROC
AFIB	0.8935	0.8993	0.8964	0.9691	0.9827
GSTV	0.9067	0.9351	0.9207	0.9731	0.9879
SB	0.9817	0.9895	0.9856	0.9891	0.9984
SR	0.9776	0.9184	0.9471	0.9952	0.9854
Total	0.9399	0.9356	0.9374	0.9816	0.9886

findings confirm the model's robustness and its capacity to generalize well across multiple cardiac conditions.

According to Fig. 12, the confusion matrix analysis on the Chapman dataset confirms that the student model trained with KD demonstrates excellent class-wise performance. The model achieved near-perfect classification for the SB class (98.95%) and exhibited strong recognition capabilities for GSTV (93.51%) and SR (91.84%). Although slightly lower, the AFIB class was also accurately predicted, with a true positive rate of 89.93%, indicating the model's effectiveness in detecting atrial fibrillation patterns. The training history presented in Fig. 12 highlights the stable and efficient learning behavior of the student model. Both the training and validation loss curves show a rapid decrease during the initial epochs, eventually converging to low and stable values (~0.15 and ~0.20, respectively), indicating effective convergence without overfitting. Similarly, the accuracy curves demonstrate consistent improvements, with the training accuracy reaching approximately 97% and validation accuracy stabilizing around 95%. The minimal gap between the training and validation performance further confirms the model's strong generalization ability. Additionally, the high accuracy achieved on both datasets, along with the small gap between them, reflects the model's strong generalization capability to unseen data. Minor fluctuations in validation accuracy are expected and do not adversely affect the overall performance.

To further evaluate the effectiveness of the proposed KD framework, extensive comparison was conducted on the PTB-XL dataset. The proposed model, which integrates MobileNetV2 and ResNet-18 enhanced with SENet and is trained under the guidance of a TA model via KD, demonstrated superior performance compared to several state-of-the-art approaches, as summarized in Table 8. The proposed student model (with KD) achieved a precision of 80.74%, recall of 74.33%, and an F1-score of 76.22%. Additionally, the model attained an accuracy of 85.45%, specificity of 95.69%, and AUROC of 0.9653, indicating a strong balance between sensitivity and specificity, crucial for handling the imbalanced nature of ECG classification tasks.

Compared to earlier works, the proposed model consistently outperformed the method by Zhang et al. [47], which reported a precision of 74.2% and an F1-score of 63.3%, along with lower accuracy and AUROC values. Similarly, Shi et al. [8] achieved an accuracy of 77.06%, specificity of 91.99%, and AUROC of 89.49%, all below those attained by the proposed model. Other approaches such as Strodthoff et al. [9] also showed inferior performance across most metrics.

More complex models like MT-MVT-Net [15] and MT-DSVS-Net [15] yielded F1-scores of 74.46% and 73.52%, respectively, with accuracies around 65%, considerably lower than the proposed model's results despite having significantly larger parameter counts (1.98 M and 0.40 M parameters). Our lightweight model, with only 101.6 K parameters, achieves comparable or better performance, highlighting the efficiency of the proposed HMT-KD framework approach.

To further validate the performance of the proposed KD framework, comprehensive experiments were conducted on the Chapman dataset. As shown in Table 9, The proposed student model reached a precision of 93.99%, recall of 93.56%, and F1-score of 93.74%, along with an accuracy of 94.42%, specificity of 98.16%, and an AUROC of 0.9886, all while maintaining a highly compact architecture with only 101 K parameters.

In comparison, Qiang et al. [15] reported an F1-score of 96.64% and an accuracy of 97.09% using MT-MVT-Net, which are slightly higher than the proposed model. However, their method requires 0.40 M parameters, nearly four times more than our lightweight design. Similarly, the ST-REGE model by Zhang et al. [47] achieved a precision of 87.9% and recall of 88.2%, resulting in a lower F1-score (88.0%) and AUROC (97.2%) than the proposed model, while also being approximately five times heavier (534 K parameters).

The model by Shi et al. [8] achieved a recall of 94.47%, accuracy of 95.10%, and AUROC of 99.43%. However, due to the absence of precision and F1-score values, a direct comparison is limited. The MobileNet model [36] yielded an F1-score of 94.1% and AUROC of 98.8%, yet lacks complete metric reporting. While it benefits from an ultra-lightweight design with just 9092 parameters, the proposed model maintains similar efficiency with more balanced and fully reported performance.

Moreover, Liu et al. [48] reported precision, recall, and F1-score values close to the proposed model (94.68%, 93.93%, and 94.25%, respectively), though their parameter count was not specified. The

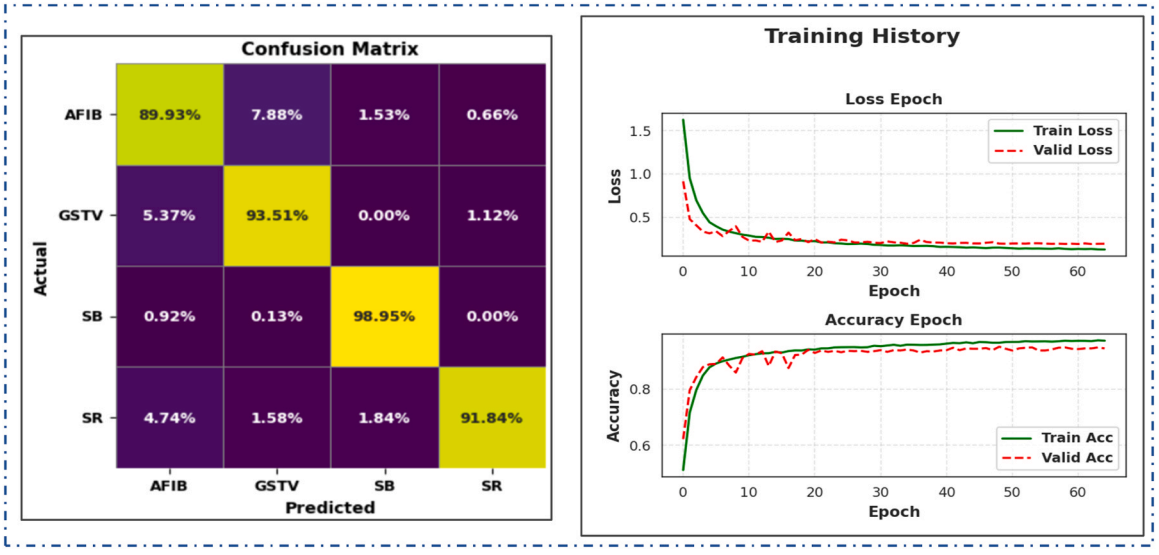


Fig. 12. Confusion matrix for the student model across five ECG classes on the Chapman dataset, and training and validation curves of the student model (with KD) on the Chapman dataset.

Table 8
The comparison of the proposed student model, which has been trained using the Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework, on the PTB-XL dataset with other existing methods.

Models	Precision	recall	F1_score	Accuracy	Specificity	AUROC	Params No.
ST-REGE [47]	0.742	0.582	0.633	0.766	_	0.891	41.19 K
GCL + CE [8]	_	0.5715	_	0.7706	0.9199	0.8949	
SSRL [48]	0.7128	0.5923	0.6169	_	_	0.9046	
MT-MVT-Net [15]	_	_	0.7446	0.6546	_	0.9381	1.98(M)
MT-DSVS-Net [15]	_	_	0.7352	0.6450	_	0.9313	0.40(M)
MobileNet_Teacher [36]	_	_	0.689	0.801	_	0.874	1.642(M)
MobileNet_Student [36]	_	_	0.661	0.781	_	0.882	9092
Ensemble [9]	_	_	_	_	_	0.934	
DConv-LSTM [27]	_	_	0.6883	0.6930	_	_	
CNN-BiLSTM [26]	0.7989	0.7308	0.7607	0.8984	0.9348	0.9351	292k
Multimodal model [49]						0.9206	17.98 M
Student (With KD)	0.8074	0.7433	0.7622	0.8545	0.9569	0.9653	101.600

Table 9
The comparison of the proposed student model trained using the Hierarchical Multi-Teacher Knowledge Distillation (HMT-KD) framework, on the Chapman dataset.

Models	Precision	recall	F1_score	Accuracy	Specificity	AUROC	Params No.
MT-MVT-Net [15]	-	-	96.64	97.09	-	98.98	0.40 M
ST-REGE [47]	0.879	0.882	0.880	0.897	-	0.972	534.71 K
GCL + CE [8]	-	0.9447	-	0.9510	0.9840	0.9943	
MobileNet_Teacher [36]			0.957	0.962		0.990	1.642(M)
MobileNet _Student [36]	-		0.941	0.949	-	0.988	9092
SSRL [48]	0.9468	0.9393	0.9425	-	-	0.9943	
CNN-BiLSTM [26]	0.9762	0.9702	0.9732	0.9880	0.9929	0.9975	292k
LFB [50]						0.9934	
JAMC [51]			0.9412	0.9492		0.9910	
Student (With KD)	0.9399	0.9356	0.9374	0.9442	0.9816	0.9886	101.343

CNN-BiLSTM model by Ayano et al. [26] demonstrated impressive performance with an F1-score of 97.32% and AUROC of 99.75%. However, it requires 292 K parameters, making it nearly three times heavier than the proposed model.

Overall, these results confirm that the proposed student model, through an effective HMT-KD framework process, achieves competitive or superior classification performance while maintaining remarkable computational efficiency. This makes it highly suitable for deployment in portable and resource-constrained clinical environments where both accuracy and efficiency are critical.

5. Conclusion

This study introduced a HMT-KD framework for ECG classification, integrating both response- and feature-level supervision. The proposed approach enables effective knowledge transfer from multiple teacher networks to a compact student model through a structured two-stage process. Initially, high-capacity teacher models distill their knowledge into a TA, which acts as an intermediary to smooth the transfer and reduce learning complexity. Subsequently, the TA and the primary teachers jointly guide the lightweight student model using both final outputs and intermediate features. Unlike conventional single-teacher or shallow KD strategies, this architecture captures a broader and more diverse representation of ECG characteristics, which improves the generalizability and robustness of the student network. The introduction of weighting based on specificity further tailors the learning process to clinical relevance by minimizing false positives in healthy cases.

Experimental evaluations on two benchmark datasets, PTB-XL and Chapman, demonstrated that the student model achieved high diagnostic accuracy (85.45% and 94.42%, respectively) and strong AUROC values (96.53% and 98.86%), while using more than 100 times fewer parameters than the ResNet-18 +SENet teacher network. These results highlight the framework's potential for real-time and resource-constrained medical applications, particularly in wearable ECG monitoring systems. Overall, the proposed method offers a scalable and practical solution for deploying deep learning models in clinical environments where computational efficiency is essential. Future research may extend this work by incorporating adaptive feature-level transfer strategies, temporal attention mechanisms, or patient-specific personalization to further enhance detection performance and clinical integration.

CRediT authorship contribution statement

Abdolhossein Fathi: Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Nasim Beigzadeh:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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